AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 14 · STUDENT LESSON

Planning and action selection

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Check the proposal. Keep the boundary. Require the evidence.

Select actions within a declared read-only task and verify each step's prerequisites before a supported answer.

Keep one habit: Task/plan contract → proposal comparison → tower-card picture → permission/prerequisite probes → executable audit → independent drafting-only case.

MAIN LESSON · SECTION 1

Treat a plan as a proposal, then check every action

> A useful-looking next step still needs permission and its prerequisites.

Learning goals

I can explain the **mechanism** and terminology of **plan**ning and **action selection**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve state guards and action limits

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your ex**plan**ation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall State machines for bounded agents. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain **plan**ning and **action selection** using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **plan**ning and **action selection**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Separate permission from evidence and sequence

> All required conditions must hold; one cannot substitute for another.

Begin with the room-helper proposal

A read-only room helper can report information without booking anything. If a proposed plan says "retrieve, book, then answer, " calling booking helpful does not authorize it. First check the whole proposal against the task boundary. Then, for a separately valid plan, check the input and evidence required by each step. A plan cannot create missing authority or turn absent evidence into a supported fact.

Define planning and action selection

Planning proposes steps; execution must still validate permissions and **prerequisite**s for each step. Re**plan**ning cannot invent authority.

Complete local task, plan and evidence contract

Main local planning contract	Complete declared rule
Task	Read-only room reporting
Accepted local order	validate → retrieve → verify → answer

Main local planning contract	Complete declared rule
Excluded operations	Booking, writing records, sending messages and other undeclared actions
Input fixture	room_id B
Approved evidence fixture	ROOM-R: room B, capacity 12, approved flag supplied
Missing evidence	Return review before verification/answering; do not invent a capacity
Whole-plan preflight	Reject forbidden actions or a missing/wrong order before simulated work
Runtime checks	Each permitted step still needs its input/evidence prerequisites
Scope	One deliberately fixed local protocol, not a universal planner or real external execution

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible ex**plan**ation as a measured outcome.

Audit a full proposal before trusting its rationale

> The expanded local bad-plan fixture includes validation and verification so the booking violation is unambiguous.

Original read-only order and fallback

For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review. Reject write or send steps before they run.

Complete unsafe, safe, missing and wrong-order paths

Reject the unsafe proposal, then audit a separately safe plan

PROPOSAL OR PATH	CHECK	OUTCOME
validate → retrieve → book → verify → answer	book is outside read-only scope	Reject entire proposed plan; no simulated stages run
validate → retrieve → verify → answer	Allowed order, valid input, matching approved fixture	Answer B/capacity12 with source ROOM-R
Same safe plan, source absent	Retrieve has no approved evidence	Review after retrieve; no verify or answer
validate → retrieve → answer → verify	Answer precedes required evidence verification	Reject wrong order before work

The first proposal is not silently repaired by deleting an action. The safe reporting plan is a separate plan checked under the same task authority.

State permissions and prerequisites before execution

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Planning and action selection — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Generated plan retrieves a room, books it and then answers. It calls booking necessary for helpfulness.	For read-only reporting, allowed plan is validate -> retrieve -> verify -> answer. If retrieval lacks evidence, route to review. Reject write or send steps before they run.	Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if evidence is insufficient.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change the evidence without inventing permission

> Keep the task boundary fixed when selecting a next action.

Current task and response

- Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the newly supplied approved card

- Change: Supply an approved card giving height 24 metres.

▼ **Check after predicting**

Prediction: The bot can retrieve it and answer 24 metres with attribution.

Reason: The missing factual **input** is now available within the allowed collection.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Choose an allowed fallback when evidence is missing

> Conditional answering authority must already be declared before a new source arrives.

Main generated proposal and task-boundary conclusion

Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness.

Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

Keep the reasoning's scope

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate tower task and complete conditional permissions

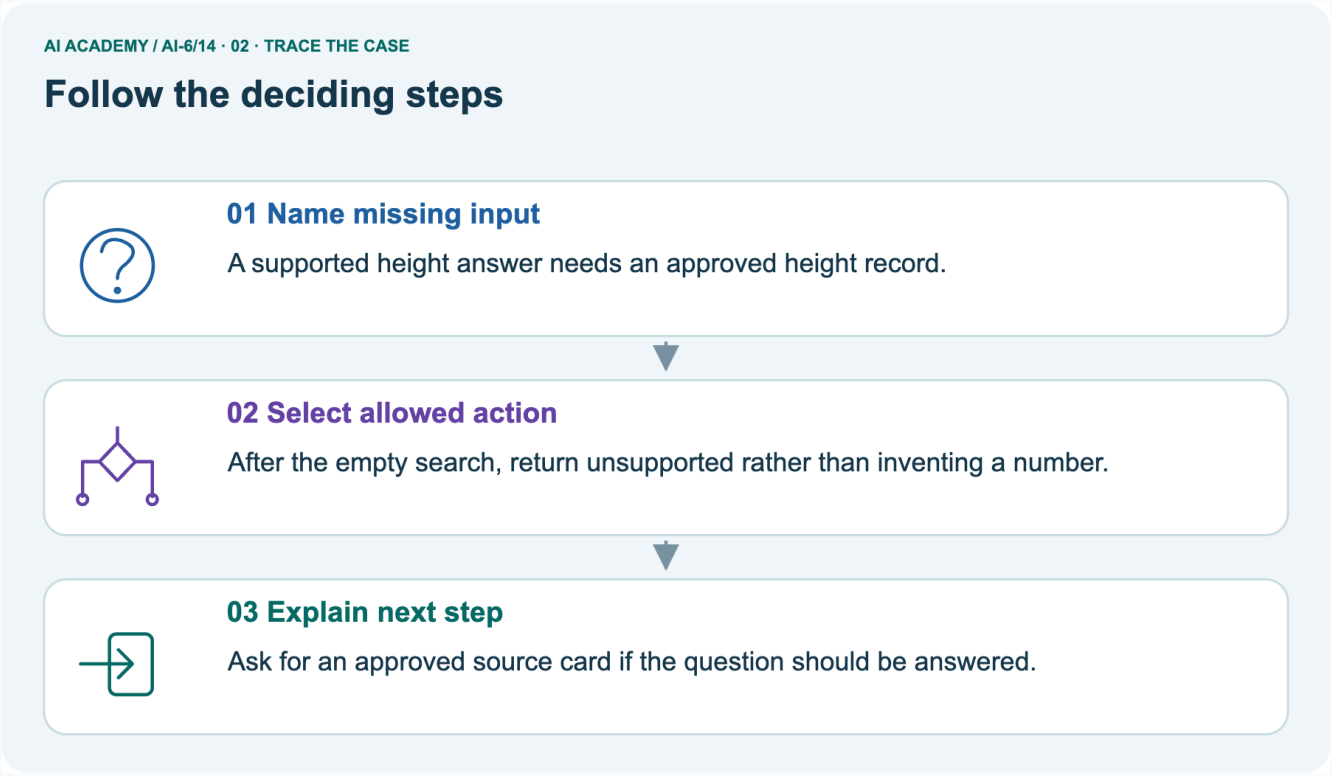
Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only.

Search finds no height card. Allowed actions: search, clarify, return unsupported, or answer from an approved card when its supporting record is present. Unapproved web access is excluded.

The main case is read-only room reporting; the picture is a separate tower-height task. Its declared actions include search, clarify, return unsupported, and conditionally answer from an approved supporting card. With no height card, return unsupported and ask for an approved card; unapproved web access stays excluded.

Supplying an approved card that states 24 metres makes that pre-authorized answer possible with attribution. The new fact satisfies an evidence prerequisite; it does not grant an extra action. This is fictional card evidence, not independently verified tower history.

Trace absent evidence and an authorized next step



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name missing input	A supported height answer needs an approved height record.
2. Select allowed action	After the empty search, return unsupported rather than inventing a number.
3. Explain next step	Ask for an approved source card if the question should be answered.

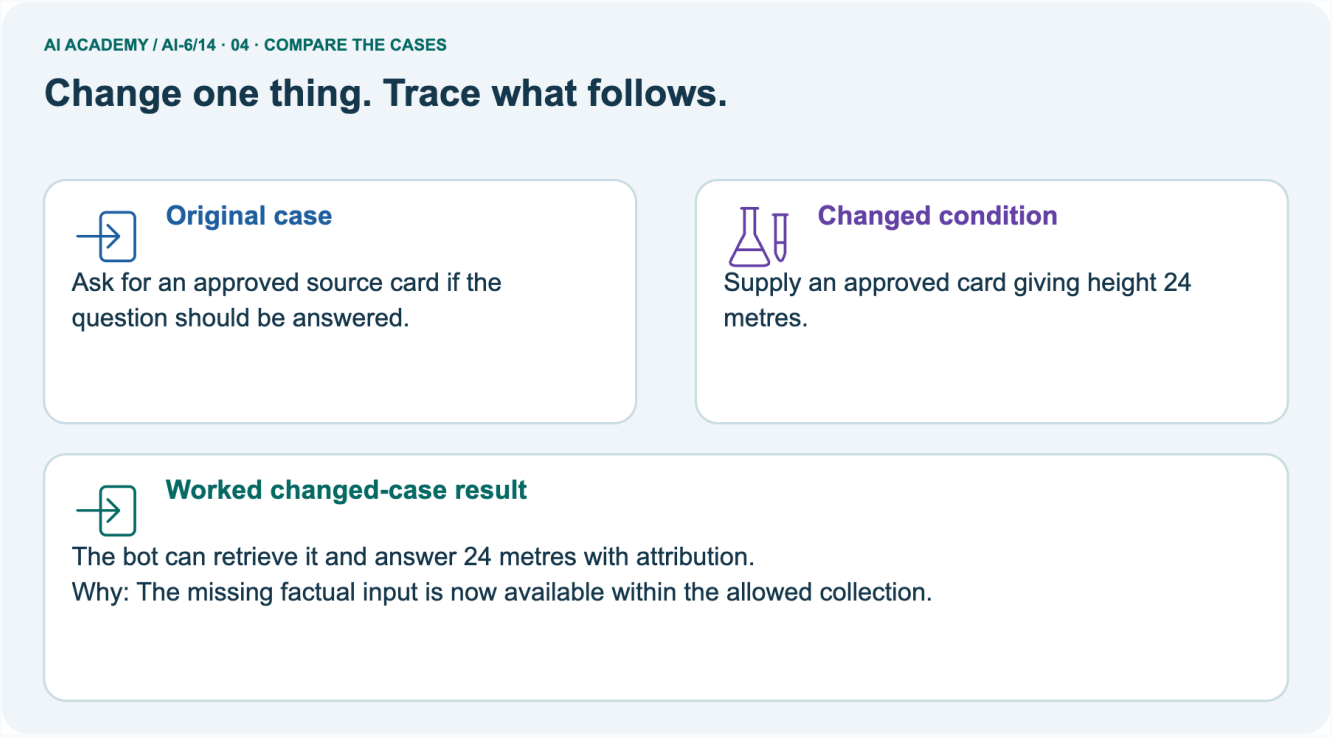
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Planning selects useful actions within available **evidence** and authority. More steps cannot create permission to access an excluded source.

Predict supplying the 24-metre card

- Change one thing: Supply an approved card giving height 24 metres. Predict the result before checking the ex**plan**ation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what became supported and what remains unverified

- Source validity still matters; the presence of a card does not independently verify its accuracy.

- Read the labelled picture: Choose a permitted **plan** when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

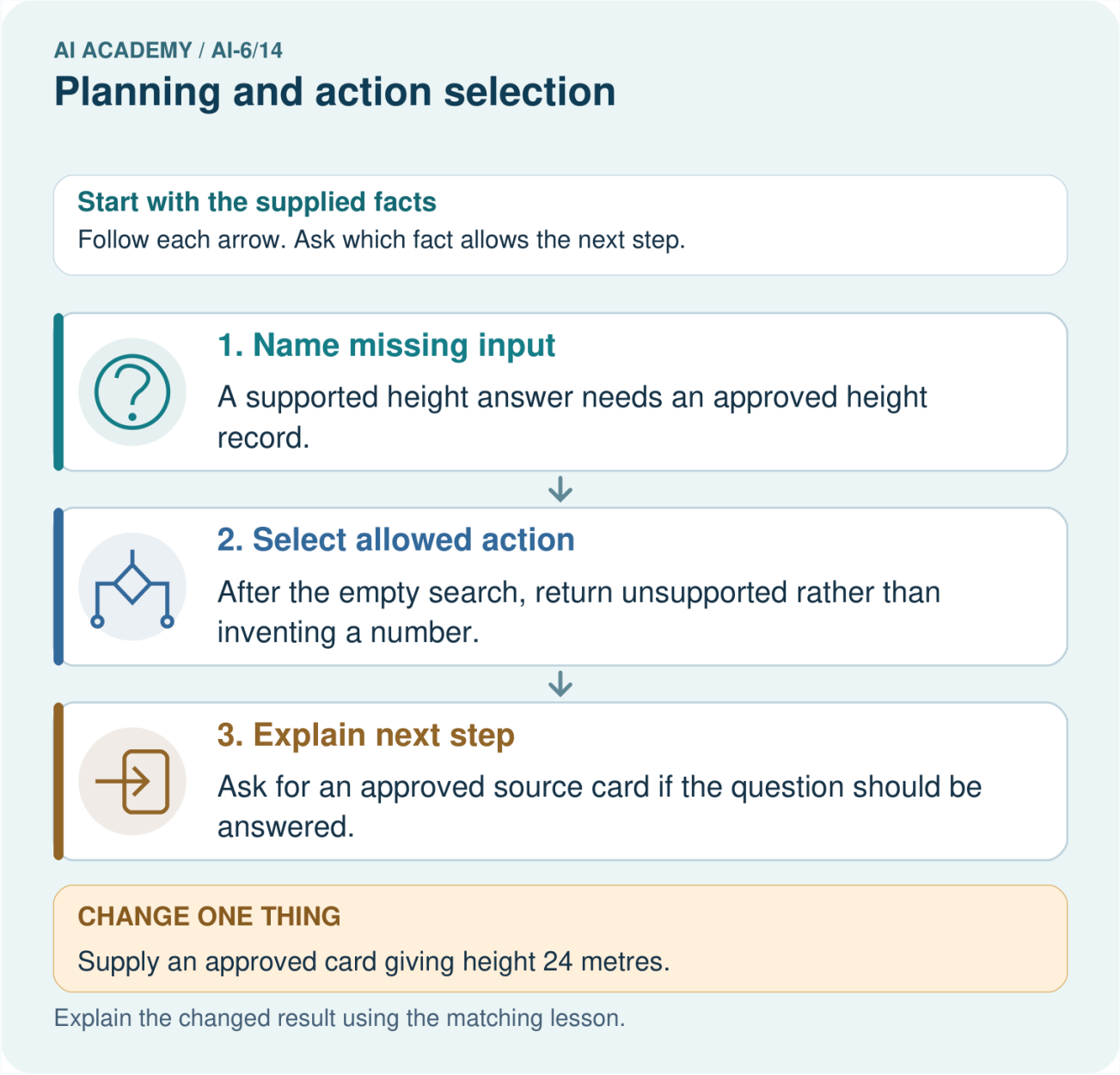
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Planning and action selection

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Check authority and prerequisites as separate conditions

> A plausible plan is neither approval nor proof that its inputs are ready.

Core planning qualification and failure analysis

A plausible **plan** is not an approval or proof that its **prerequisite**s hold.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete action, ordering and evidence probes

Plan or prerequisite probe	Expected decision
Room proposal includes book	Reject outside task authority, regardless of helpfulness rationale
Safe reporting plan with B/12/ROOM-R evidence	May produce attributed report under the local contract
Safe plan with no source	Review before unsupported answering
Answer placed before verify	Reject wrong order; plausible final wording cannot fix it
Input evidence exists but belongs to a different room	Review; source presence alone does not establish support
Tower search lacks an approved card	Unsupported/clarify under the picture policy; no excluded web access
Approved tower card states 24 metres	Conditional permitted answer becomes evidence-supported, not new authority

Plan or prerequisite probe	Expected decision
A prerequisite is satisfied	Still check action permission; prerequisites do not expand scope
Local execution ledger contains steps	Shows simulated control flow, not a real booking, message or model plan

Guided explanation and repair

- A report omits this qualification: A plausible **plan** is not an approval or proof that its **prerequisite**s hold. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise naming the violated permission or missing prerequisite

> Explain why the next action follows from the task and current evidence.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example "**Plan**ning and **action selection**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own ex**plan**ation.

Individual plan-versus-approval check

- Use the worked case for **Plan**ning and **action selection**. Proposed ex**plan**ation: “The worked result establishes every future case”. Decide whether this ex**plan**ation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Tower picture response

- Choose a permitted **plan** when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

> **My explanation using a pictured step and the given facts**
>
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Reserve independent drafting-only message task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete proposal and runtime-fallback audit

> Retain rejected proposals, permitted-stage traces and evidence failures together.

Current booking-in-read-only-plan investigation

Generated **plan** retrieves a room, books it and then answers. It calls booking necessary for helpfulness. Show the intermediate steps and give the conclusion with its scope.

Complete runnable proposal and source checks

```
"""Local read-only plan audit; no model planning, booking, messaging or web calls."""
SAFE=['validate','retrieve','verify','answer']
PERMITTED=frozenset(SAFE)
SOURCE={'source_id':'ROOM-R','room_id':'B','capacity':12,'approved':True}

def run_plan(plan,request,source,events):
    if type(plan) is not list or any(type(step) is not str for step in plan):
        return {'status':'rejected_plan','reason':'invalid plan structure'}
    # Preflight rejects the entire unsafe proposal before any simulated step.
    if any(step not in PERMITTED for step in plan):
        return {'status':'rejected_plan','reason':'operation outside read-only scope'}
    if plan!=SAFE:
        return {'status':'rejected_plan','reason':'required order or stage missing'}
    validated=False;retrieved=None;verified=False
    for action in plan:
        if action not in PERMITTED:return {'status':'rejected_plan'}
        events.append(action)
        if action=='validate':
            if (type(request) is not dict or set(request)!={'room_id'}
                or type(request['room_id']) is not str or not request['room_id']):
                return {'status':'invalid_request'}
            validated=True
        elif action=='retrieve':
            if not validated:return {'status':'review','reason':'missing validation'}
            retrieved=source  # local injected fixture, not a network lookup
            if retrieved is None:return {'status':'review','reason':'missing approved
evidence'}
            continue
        elif action=='verify':
```

```
            if (type(retrieved) is not dict or
                set(retrieved)!={'source_id','room_id','capacity','approved'} or
                retrieved['approved'] is not True or
                retrieved['room_id']!=request['room_id'] or
                type(retrieved['source_id']) is not str or not retrieved['source_id'] or
                type(retrieved['capacity']) is not int or retrieved['capacity']<0):
                return {'status':'review','reason':'evidence verification failed'}
            verified=True
        elif action=='answer':
            if not verified:return {'status':'review','reason':'missing verified
evidence'}
            return {'status':'answered','room_id':retrieved['room_id'],
                    'capacity':retrieved['capacity'],'source_id':retrieved['source_id']}
    return {'status':'review'}

unsafe=['validate','retrieve','book','verify','answer']
unsafe_events=[]
rejected=run_plan(unsafe,{'room_id':'B'},SOURCE,unsafe_events)
assert rejected['status']=='rejected_plan' and unsafe_events==[]
safe_events=[]
answer=run_plan(list(SAFE),{'room_id':'B'},SOURCE,safe_events)
assert answer=={'status':'answered','room_id':'B','capacity':12,'source_id':'ROOM-R'}
assert safe_events==SAFE
missing_events=[]
missing=run_plan(list(SAFE),{'room_id':'B'},None,missing_events)
assert missing['status']=='review' and missing_events==['validate','retrieve']
wrong_events=[]
wrong=run_plan(['validate','retrieve','answer','verify'],
{'room_id':'B'},SOURCE,wrong_events)
assert wrong['status']=='rejected_plan' and wrong_events==[]
print('Unsafe proposal rejected before all simulated work:',rejected,unsafe_events)
print('Separately safe read-only plan:',answer,safe_events)
print('Missing evidence runtime fallback:',missing,missing_events)
print('Wrong-order proposal:',wrong,wrong_events)
# Separate pictured tower-card policy permits conditional supported answering too.
def tower_reply(card):
    if card is None:return {'status':'unsupported','next':'request an approved card'}
    if card=={'source_id':'HEIGHT-CARD','height_m':24,'approved':True}:
        return {'status':'answered','height_m':24,'source_id':'HEIGHT-CARD'}
    return {'status':'review'}
assert tower_reply(None)['status']=='unsupported'
assert tower_reply({'source_id':'HEIGHT-CARD','height_m':24,'approved':True})
['height_m']==24
print('Tower without/with supplied approved
card:',tower_reply(None),tower_reply({'source_id':'HEIGHT-
CARD','height_m':24,'approved':True}))
```

```
print('Fixtures only: the approved flag is supplied, not independent source verification;
no external action executes.')
```

Expected paths, fixed-protocol scope and limits

1. Predict plan status and event list before running or tracing. The unsafe plan contains book and is rejected before simulated work. A separately safe fixed-order plan reports room B/capacity 12/source ROOM-R; missing evidence stops after retrieve with review. Wrong-order answer-before-verify is rejected at preflight.

   The local contract deliberately accepts one exact safe sequence; a more flexible planner would need separately declared rules. Runtime checks use supplied local evidence and a mock approved flag, not independent verification. No model planning, real tool call, booking, message delivery or web search occurs. The tower example separately includes conditional supported answering in its original declared scope.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: inspect a drafting-only plan that tries to send

> Use both task authority and prerequisite order in the new scenario.

Complete independent message-plan task

Plan validates **input** then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Attempt this without hints

Plan validates **input** then sends a message before checking recipient approval. The task authorises drafting only. Locate violation.

> **My independent procedure, result and limitation**
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> | |

Record support used: none, hint or teacher ex**plan**ation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the task boundary beside the proposed and accepted plans

> Keep each action’s permission, prerequisite and observed local outcome.

Project task and artifact

Add a labelled **plan**ning and **action selection evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **plan**ning and **action selection**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with an authorized next action and a supported claim

> More planning cannot create permission or missing source facts.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **plan**ning and **action selection**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Plan	A proposed ordered set of steps
Prerequisite	A condition required before a step
Action selection	Choosing a permitted operation under the current state

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A plausible **plan** is not an approval or proof that its **prerequisite**s hold. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Planning and action selection — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Generated plan retrieves a room, books it and then answers. It calls booking necessary for helpfulness.	For read-only reporting, allowed plan is validate -> retrieve -> verify -> answer. If retrieval lacks evidence, route to review. Reject write or send steps before they run.	Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if evidence is insufficient.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: **Plan** around missing **evidence**. A bot must answer a tower-height question using approved cards only.

By the end, I can	How I will show it
Explain the mechanism and terminology of planning and action selection	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- **Plan** around missing **evidence**. A bot must answer a tower-height question using approved cards only.
- Which labelled step matches this detail? Ask for an approved source card if the question should be answered.
- 1. Select allowed action
- 2. Explain next step
- 3. Name missing **input**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported ex**plan**ation: Explain next step Ask for an approved source card if the question should be answered. **Plan**ning selects useful actions within available **evidence** and authority.

More steps cannot create permission to access an excluded source.

Changed-case question and optional hint

- Changed-case practice: Supply an approved card giving height 24 metres. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The bot can retrieve it and answer 24 metres with attribution.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported ex**plan**ation: The bot can retrieve it and answer 24 metres with attribution. The missing factual **input** is now available within the allowed collection.

Repair and retain the record

Review State machines for bounded agents (ai-6/13). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram.

Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Review Testing model-dependent behaviour (ai-6/8). Goal: Explain the **mechanism** and terminology of testing model-dependent behaviour.

Short practice: Score "Friday in room B" and "Friday in room D" against both mandatory criteria. Would a correct 25-word response pass, and why?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

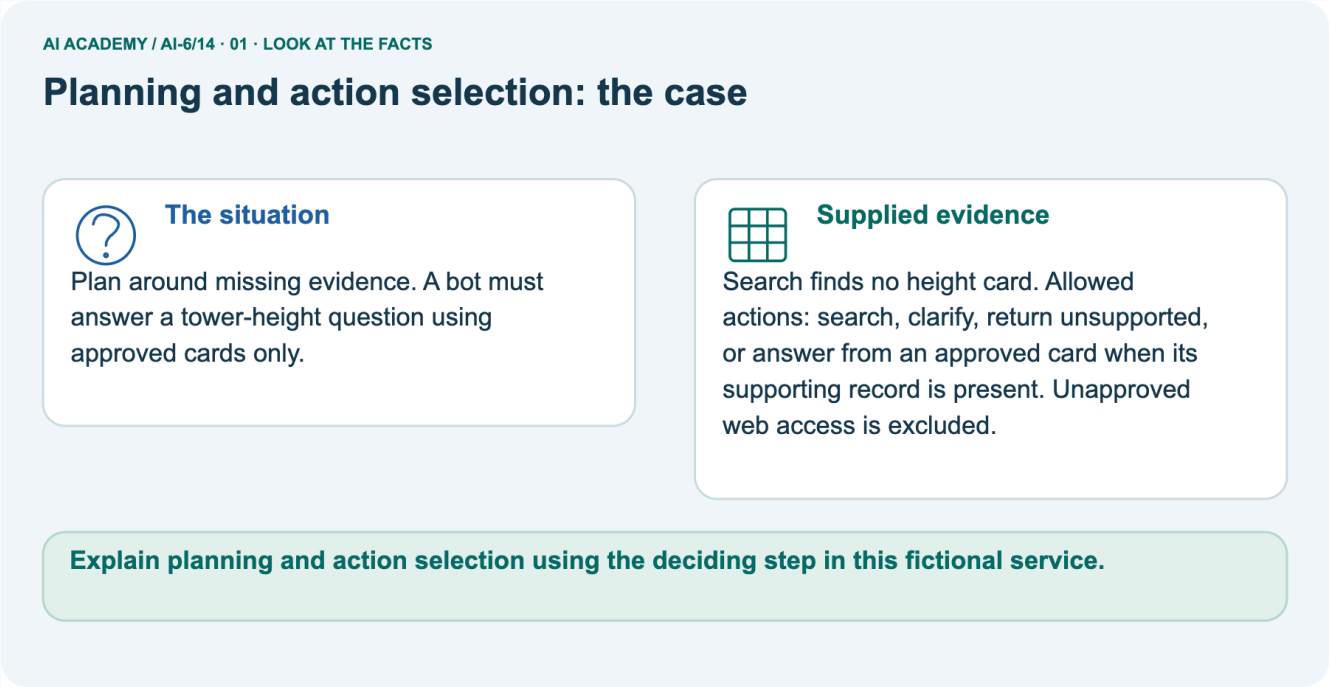
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Planning and action selection: the case

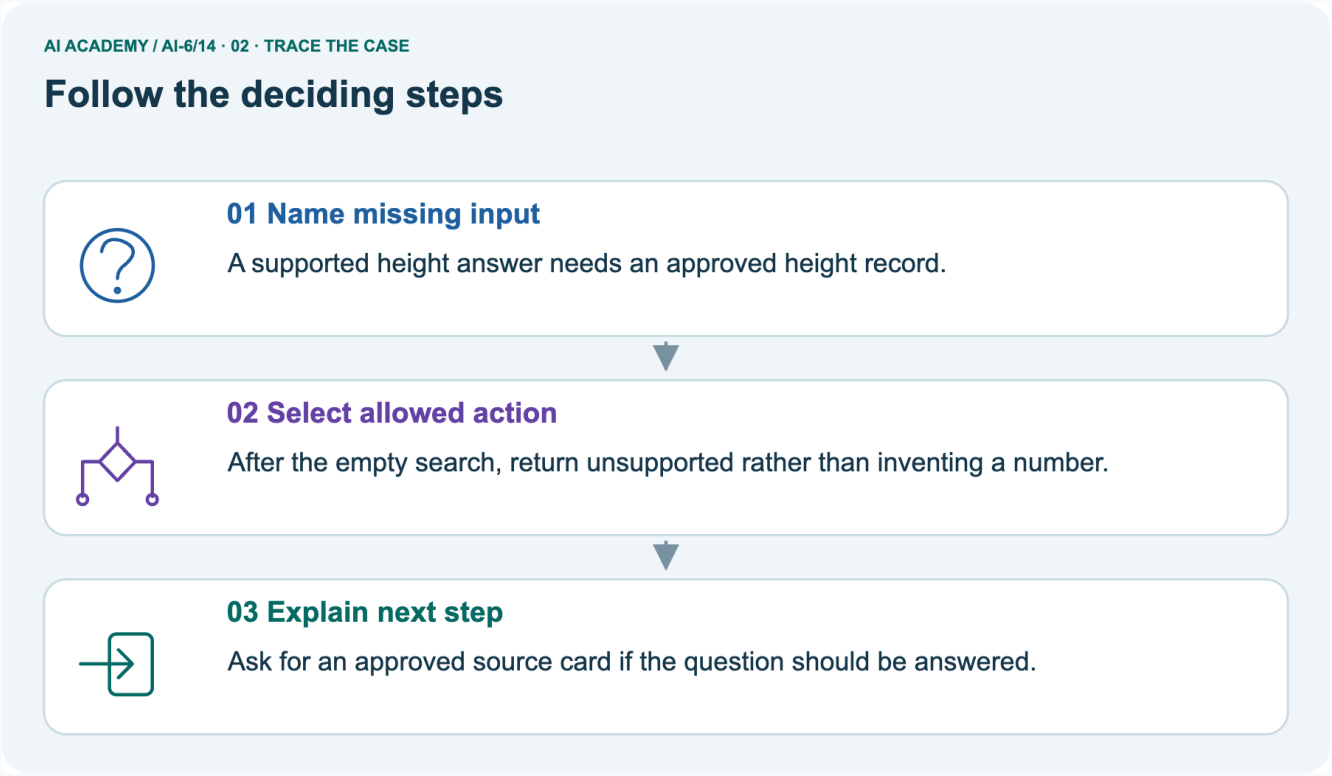


Follow the labelled picture with the matching explanation. Open the vector diagram

Plan around missing **evidence**. A bot must answer a tower-height question using approved cards only. Supplied facts: Search finds no height card. Allowed actions: search, clarify, return unsupported, or answer from an approved card when its supporting record is present. Unapproved web access is excluded.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

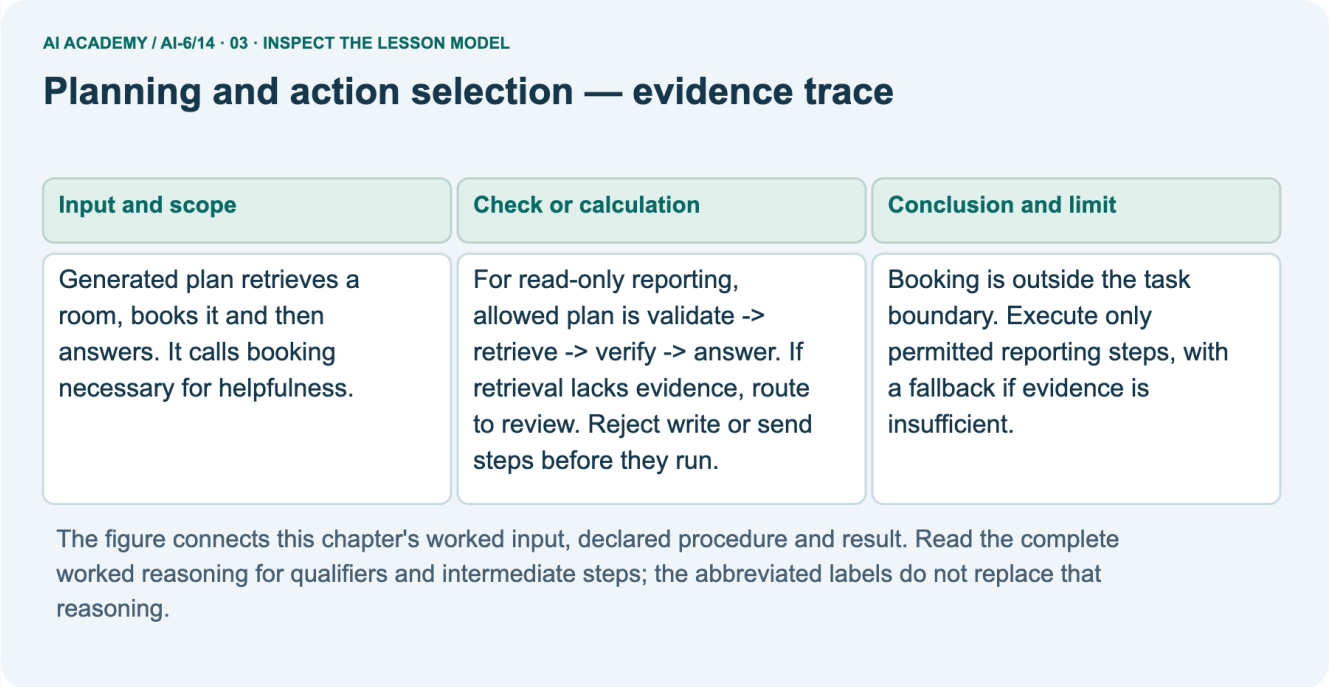


Follow the labelled picture with the matching explanation. Open the vector diagram

Name missing **input**: A supported height answer needs an approved height record. → Select allowed action: After the empty search, return unsupported rather than inventing a number. → Explain next step: Ask for an approved source card if the question should be answered.

Point to each numbered stage and explain which recorded **input** it uses.

03. Planning and action selection — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

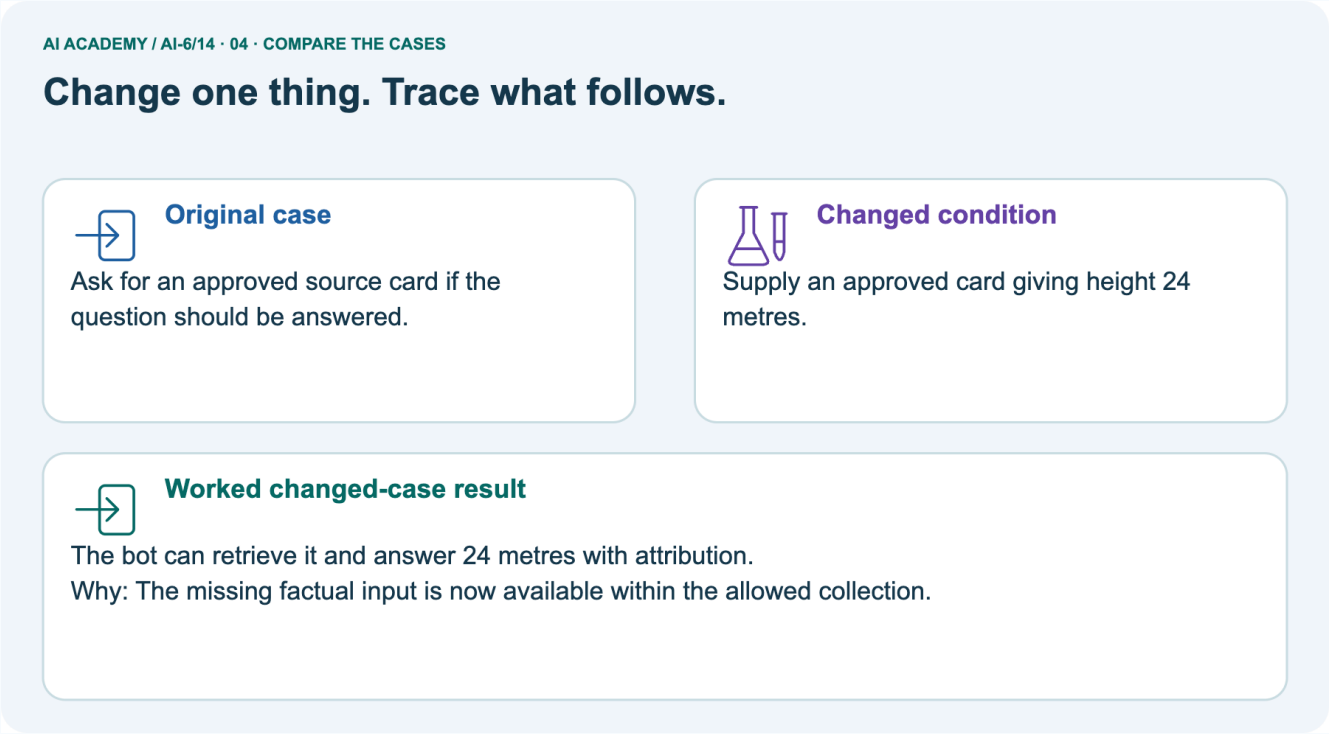
Planning and **action selection** — **evidence** trace. Generated **plan** retrieves a room, books it and then answers.

It calls booking necessary for helpfulness. | For read-only reporting, allowed **plan** is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review.

Reject write or send steps before they run. | Booking is outside the task boundary. Execute only permitted reporting steps, with a fallback if **evidence** is insufficient.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



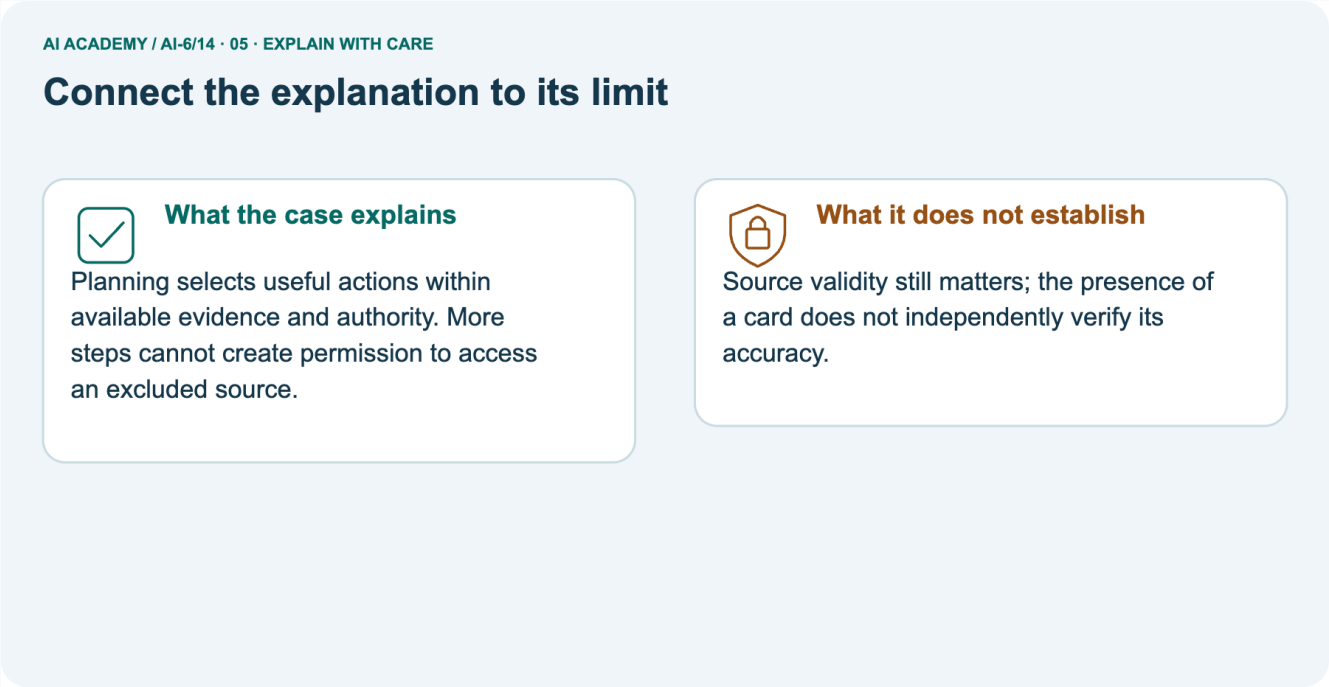
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Ask for an approved source card if the question should be answered. Changed condition: Supply an approved card giving height 24 metres.

Expected result: The bot can retrieve it and answer 24 metres with attribution. Reason: The missing factual **input** is now available within the allowed collection.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



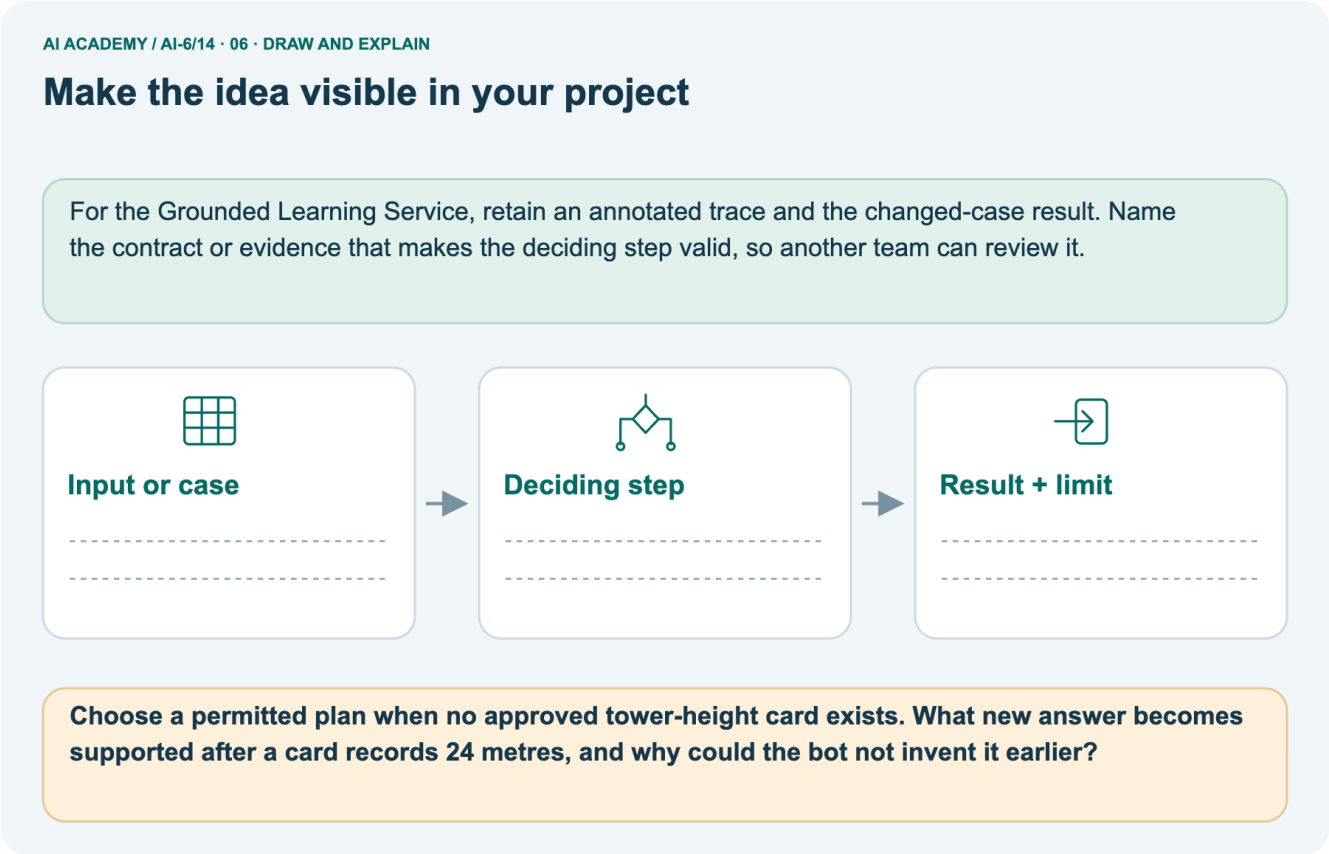
Follow the labelled picture with the matching explanation. Open the vector diagram

Planning selects useful actions within available **evidence** and authority. More steps cannot create permission to access an excluded source.

Limit: Source validity still matters; the presence of a card does not independently verify its accuracy.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Choose a permitted **plan** when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Planning and action selection: the case

02. Follow the deciding steps

03. Planning and action selection — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.